Corso di Laurea in Fisica ; Corso di Struttura della Materia
Alcune formule e dati utili per la risoluzione dei problemi

Fundamental Physical Constants — Frequently used constants

Quantity	Symbol	Value	Unit	Relative std. uncert. u_r
speed of light in vacuum	c, c_0	299 792 458	$m\ s^{-1}$	(exact)
magnetic constant	μ_0	$4\pi \times 10^{-7}$	$N\ A^{-2}$	
		$= 12.566\,370\,614... \times 10^{-7}$	$N\ A^{-2}$	(exact)
electric constant $1/\mu_0 c^2$	ϵ_0	$8.854\,187\,817... \times 10^{-12}$	$F\ m^{-1}$	(exact)
Newtonian constant of gravitation	G	$6.674\,28(67) \times 10^{-11}$	$m^3\ kg^{-1}\ s^{-2}$	1.0×10^{-4}
Planck constant	h	$6.626\,068\,96(33) \times 10^{-34}$	J s	5.0×10^{-8}
$h/2\pi$	$\hbar$	$1.054\,571\,628(53) \times 10^{-34}$	J s	5.0×10^{-8}
elementary charge	e	$1.602\,176\,487(40) \times 10^{-19}$	C	2.5×10^{-8}
magnetic flux quantum $h/2e$	Φ_0	$2.067\,833\,667(52) \times 10^{-15}$	Wb	2.5×10^{-8}
conductance quantum $2e^2/h$	G_0	$7.748\,091\,7004(53) \times 10^{-5}$	S	6.8×10^{-10}
electron mass	m_e	$9.109\,382\,15(45) \times 10^{-31}$	kg	5.0×10^{-8}
proton mass	m_p	$1.672\,621\,637(83) \times 10^{-27}$	kg	5.0×10^{-8}
proton-electron mass ratio	m_p/m_e	$1836.152\,672\,47(80)$		4.3×10^{-10}
fine-structure constant $e^2/4\pi\epsilon_0\hbar c$	α	$7.297\,352\,5376(50) \times 10^{-3}$		6.8×10^{-10}
inverse fine-structure constant	α^{-1}	$137.035\,999\,679(94)$		6.8×10^{-10}
Rydberg constant $\alpha^2 m_e c/2h$	R_∞	$10\,973\,731.568\,527(73)$	m^{-1}	6.6×10^{-12}
Avogadro constant	N_A, L	$6.022\,141\,79(30) \times 10^{23}$	mol^{-1}	5.0×10^{-8}
Faraday constant $N_A e$	F	$96\,485.3399(24)$	$C\ mol^{-1}$	2.5×10^{-8}
molar gas constant	R	$8.314\,472(15)$	$J\ mol^{-1}\ K^{-1}$	1.7×10^{-6}
Boltzmann constant R/N_A	k	$1.380\,6504(24) \times 10^{-23}$	$J\ K^{-1}$	1.7×10^{-6}
Stefan-Boltzmann constant $(\pi^2/60)k^4/\hbar^3 c^2$	σ	$5.670\,400(40) \times 10^{-8}$	$W\ m^{-2}\ K^{-4}$	7.0×10^{-6}
Non-SI units accepted for use with the SI				
electron volt: (e/C) J	eV	$1.602\,176\,487(40) \times 10^{-19}$	J	2.5×10^{-8}

Bohr radius: $a_0 = \dfrac{4\pi\varepsilon_0 \hbar^2}{m_e e^2} = 0.0529177\ \text{nm}$; **Bohr magneton:** $\mu_B = \dfrac{e \cdot \hbar}{2 \cdot m_e} = 5.7883 \times 10^{-5}\ \text{eV} \cdot \text{T}^{-1}$

Wien Displacement Constant: 2.8977685×10^{-3} m·K **;**

Funzioni d'onda di atomi idrogenoidi

$$\rho = r \cdot \left(\frac{2Z}{na_0}\right)$$

$$n=1, l=0, m=0 \Rightarrow \psi_{1,0,0}(r) = \frac{2}{\sqrt{4\pi}} \cdot \left(\frac{Z}{a_0}\right)^{3/2} \cdot \exp\left(-\frac{\rho}{2}\right)$$

$$n=2, l=0, m=0 \Rightarrow \psi_{2,0,0}(r) = \frac{1}{4\sqrt{2\pi}} \cdot \left(\frac{Z}{a_0}\right)^{3/2} \cdot (2-\rho) \cdot \exp\left(-\frac{\rho}{2}\right)$$

$$n=2, l=1, m=0 \Rightarrow \psi_{2,1,0}(r) = \frac{1}{4\sqrt{2\pi}} \cdot \left(\frac{Z}{a_0}\right)^{3/2} \cdot \rho \cdot \exp\left(-\frac{\rho}{2}\right) \cdot \cos\theta$$

$$n=2, l=1, m=\pm1 \Rightarrow \psi_{2,1,\pm1}(r) = \frac{1}{8\sqrt{\pi}} \cdot \left(\frac{Z}{a_0}\right)^{3/2} \cdot \rho \cdot \exp\left(-\frac{\rho}{2}\right) \cdot \sin\theta \cdot \exp(\pm i\phi)$$

Integrali notevoli:

$$\int x \cdot \exp[-a \cdot x]dx = -\frac{\exp[-a \cdot x]}{a^2} \cdot \{a \cdot x + 1\};$$

$$\int x^2 \cdot \exp[-a \cdot x]dx = -\frac{\exp[-a \cdot x]}{a^3} \cdot \{a^2 \cdot x^2 + 2 \cdot a \cdot x + 2\};$$

$$\int x^3 \cdot \exp[-a \cdot x]dx = -\frac{\exp[-a \cdot x]}{a^4} \cdot \{a^3 \cdot x^3 + 3 \cdot a^2 \cdot x^2 + 6 \cdot a \cdot x + 6\};$$

$$\int x^4 \cdot \exp[-a \cdot x]dx = -\frac{\exp[-a \cdot x]}{a^5} \cdot \{a^4 \cdot x^4 + 4 \cdot a^3 \cdot x^3 + 12 \cdot a^2 \cdot x^2 + 24 \cdot a \cdot x + 24\};$$

$$\int_0^\infty x^n \cdot \exp[-a \cdot x]dx = \frac{n!}{a^{n+1}}$$

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Tavola periodica degli elementi

Key:
element name
atomic number
symbol
atomic weight (mean relative mass)

hydrogen 1 **H** 1.0079																		helium 2 **He** 4.0026
lithium 3 **Li** 6.941	beryllium 4 **Be** 9.0122												boron 5 **B** 10.811	carbon 6 **C** 12.011	nitrogen 7 **N** 14.007	oxygen 8 **O** 15.999	fluorine 9 **F** 18.998	neon 10 **Ne** 20.180
sodium 11 **Na** 22.990	magnesium 12 **Mg** 24.305												aluminium 13 **Al** 26.982	silicon 14 **Si** 28.086	phosphorus 15 **P** 30.974	sulfur 16 **S** 32.065	chlorine 17 **Cl** 35.453	argon 18 **Ar** 39.948
potassium 19 **K** 39.098	calcium 20 **Ca** 40.078		scandium 21 **Sc** 44.956	titanium 22 **Ti** 47.867	vanadium 23 **V** 50.942	chromium 24 **Cr** 51.996	manganese 25 **Mn** 54.938	iron 26 **Fe** 55.845	cobalt 27 **Co** 58.933	nickel 28 **Ni** 58.693	copper 29 **Cu** 63.546	zinc 30 **Zn** 65.39	gallium 31 **Ga** 69.723	germanium 32 **Ge** 72.61	arsenic 33 **As** 74.922	selenium 34 **Se** 78.96	bromine 35 **Br** 79.904	krypton 36 **Kr** 83.80
rubidium 37 **Rb** 85.468	strontium 38 **Sr** 87.62		yttrium 39 **Y** 88.906	zirconium 40 **Zr** 91.224	niobium 41 **Nb** 92.906	molybdenum 42 **Mo** 95.94	technetium 43 **Tc** [98]	ruthenium 44 **Ru** 101.07	rhodium 45 **Rh** 102.91	palladium 46 **Pd** 106.42	silver 47 **Ag** 107.87	cadmium 48 **Cd** 112.41	indium 49 **In** 114.82	tin 50 **Sn** 118.71	antimony 51 **Sb** 121.76	tellurium 52 **Te** 127.60	iodine 53 **I** 126.90	xenon 54 **Xe** 131.29
caesium 55 **Cs** 132.91	barium 56 **Ba** 137.33	57-70 *	lutetium 71 **Lu** 174.97	hafnium 72 **Hf** 178.49	tantalum 73 **Ta** 180.95	tungsten 74 **W** 183.84	rhenium 75 **Re** 186.21	osmium 76 **Os** 190.23	iridium 77 **Ir** 192.22	platinum 78 **Pt** 195.08	gold 79 **Au** 196.97	mercury 80 **Hg** 200.59	thallium 81 **Tl** 204.38	lead 82 **Pb** 207.2	bismuth 83 **Bi** 208.98	polonium 84 **Po** [209]	astatine 85 **At** [210]	radon 86 **Rn** [222]
francium 87 **Fr** [223]	radium 88 **Ra** [226]	89-102 **	lawrencium 103 **Lr** [262]	rutherfordium 104 **Rf** [261]	dubnium 105 **Db** [262]	seaborgium 106 **Sg** [266]	bohrium 107 **Bh** [264]	hassium 108 **Hs** [269]	meitnerium 109 **Mt** [268]	ununnilium 110 **Uun** [271]	unununium 111 **Uuu** [272]	ununbium 112 **Uub** [277]		ununquadium 114 **Uuq** [289]				

*lanthanoids	lanthanum 57 **La** 138.91	cerium 58 **Ce** 140.12	praseodymium 59 **Pr** 140.91	neodymium 60 **Nd** 144.24	promethium 61 **Pm** [145]	samarium 62 **Sm** 150.36	europium 63 **Eu** 151.96	gadolinium 64 **Gd** 157.25	terbium 65 **Tb** 158.93	dysprosium 66 **Dy** 162.50	holmium 67 **Ho** 164.93	erbium 68 **Er** 167.26	thulium 69 **Tm** 168.93	ytterbium 70 **Yb** 173.04
**actinoids	actinium 89 **Ac** [227]	thorium 90 **Th** 232.04	protactinium 91 **Pa** 231.04	uranium 92 **U** 238.03	neptunium 93 **Np** [237]	plutonium 94 **Pu** [244]	americium 95 **Am** [243]	curium 96 **Cm** [247]	berkelium 97 **Bk** [247]	californium 98 **Cf** [251]	einsteinium 99 **Es** [252]	fermium 100 **Fm** [257]	mendelevium 101 **Md** [258]	nobelium 102 **No** [259]